Auditing and Learning Masked-Evidence Memory Interfaces in Latent World Models

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Abstract

Finite-context latent world models can store old evidence internally yet fail to expose it to a later decision. We study this as an interface problem rather than an architecture label. A valid memory claim must establish old-cue demand, post-context retention, frozen-host exposure, and downstream use under matched controls. We instantiate this ladder with masked-evidence sidecars on frozen DINO-WM and LeWM hosts, where PushT visual binding passes a checkpointed downstream-use gate and DINO-WM PointMaze converts retained goal state into fixed-controller execution gains. Wall exposes the key failure mode: carrier-prior memory remains high while host-output exposure collapses with cue age. We then remove the manual cue-feature handoff with a self-supervised salient patch-set JEPA objective: target frames and patches are mined without cue labels, cue crops, or key-frame annotations, and a compact slot state predicts fixed generic color/texture latents. On OGBench PointMaze-large and Cube-single this non-manual objective passes all 30 registered cells over ages 4/8/15 and five seeds: full-memory balanced accuracy is 0.92–1.00 while reset and recent-only controls stay near chance. The result is a scoped method-and-audit contribution: persistent memory in latent world models is a property of the target–carrier–host–consumer interface, not a universal property of any single recurrent, state-space, or fixed-trust module.

1 Introduction

Finite-context latent world models make visual control tractable by compressing history into a short latent context. That same compression makes memory claims ambiguous. A model can produce locally plausible next features while losing an old cue that a later decision requires; a recurrent or state-space side state can store information internally while the frozen host never exposes it; a linear readout can recover a symbol without showing that an executed action would change. These are different failure modes, but they are often collapsed into one question: does the model have memory?

We study memory as an interface property of a frozen latent host. The host is not retrained. A compact causal side state observes the legal observation/action stream, stores evidence that has left the raw context, and writes only through the frozen host interface. The audit is fail-closed: stronger claims are attempted only after earlier gates pass. The resulting claim ladder has four rungs:

1. demand: the target cannot be solved from the legal endpoint, action, or state shortcuts;
2. retention: the side state causally carries the old evidence after context expiry;
3. host exposure: the frozen predictor exposes that evidence in its output rather than hiding it inside the side state; and
4. use: a fixed consumer can convert the exposed state into an executed decision.

The paper is therefore not a claim that one carrier architecture wins. It is a claim that persistent memory must be audited as a target–carrier–host– consumer chain.

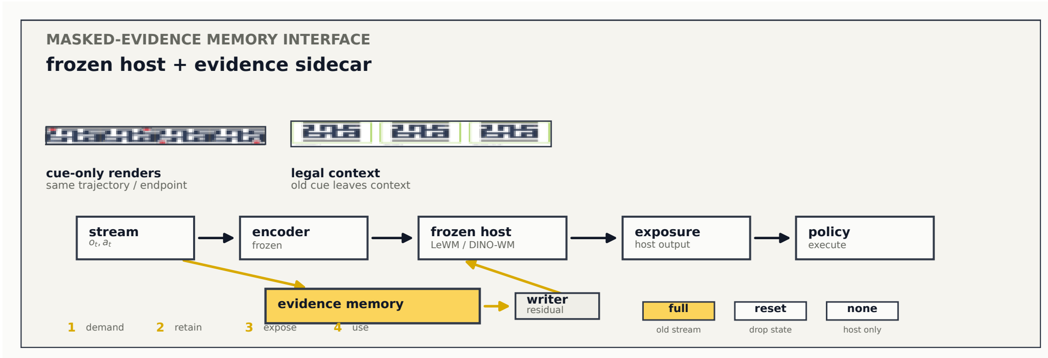

Figure 1: Masked-evidence memory interface. The top strip shows native-ratio rendered inputs: cue-only counterfactuals share trajectory, actions, and endpoint, while the legal context no longer contains the old cue. The model schematic follows a frozen encoder/host path with a causal evidence sidecar that writes through a residual host interface before exposure and use.

Contributions. This study makes four scoped contributions. First, it turns memory evaluation into a controlled interface audit with explicit demand, retention, exposure, and use gates. Second, it validates positive downstream-use cases on PushT and PointMaze while keeping the consumer boundary explicit. Third, it uses Wall and OGBench to prevent overclaiming: Wall isolates a storage-versus- exposure bottleneck, and OGBench shows that the admission protocol ports across seven environment families while a repaired PointMaze/Cube/Puzzle/Scene fixed-controller sweep tests whether the selected memory target changes native environment success without claiming native OGBench planning. Fourth, it instantiates the audit as a non-manual self-supervised memory objective: salient target frames and patch sets are mined without cue labels, cue crops, or key-frame annotations, and the slot memory predicts fixed generic color/texture latents. This objective passes all 30 registered OGBench PointMaze-large/Cube-single cells over ages 4/8/15 and five seeds.

2 Audit design

Frozen hosts and sidecar states. Each study starts from a locked finite-context latent host, either DINO-WM (Zhou et al., 2025) or LeWM (Maes et al., 2026). The encoder and predictor remain fixed. A sidecar state receives the same legal stream as the host and is evaluated under matched conditions. The full condition receives the old stream. The reset condition reinitializes memory at the legal context boundary. The no-state condition evaluates the frozen host without a persistent side state. These controls do not estimate an abstract scalar “amount of memory”; they localize where the interface stops carrying old evidence.

Masked evidence, not declared key frames. The sidecar is not given a symbolic instruction that a particular past frame will matter. Instead, the task constructs cue-only counterfactuals: the old evidence changes while the later endpoint, action stream, and other legal signals stay fixed. The target can therefore be solved only if the side state keeps the old evidence. In feature-host settings the sidecar can be evaluated both as a carrier prior and through the host output; separating these two views is essential, because an internally readable carrier is not automatically a usable world-model output.

Autonomous salient patch-set JEPA. The final OGBench method removes the remaining manual cue-feature handoff. It samples a causal rendered stream, mines a salient target frame and a set of high-saliency patches without using cue labels or cue layout, masks only that target view, and asks a compact slot state to predict stop-gradient color/texture patch latents. Labels are never used in the learning objective; they enter only after training through the same full/reset/recent-only audit. This design is intentionally not a KV cache, a recurrent hidden state, or a manually declared key-frame mechanism. The model must learn which evidence is worth carrying because the generic patch-set objective repeatedly withholds salient old observations from the legal context.

TASK-LEVEL CLAIM LEDGER

One paper result is not one memory score.

env	demand	retention	host exposure	executed use	role in paper
PushT	✓	✓	✓	✓	positive use, two hosts
PointMaze	✓	✓	✓	✓	strong navigation use
Wall	✓	✓	mixed	–	exposure bottleneck
OGBench	✓	cap	cap	✓	fixed-controller success

Figure 2: Task-level claim ledger. The paper avoids a pooled memory score: each row answers a different interface question and carries its own claim boundary. A check mark means that the corresponding rung is supported by the current artifact; a dash means the rung is intentionally not claimed.

Admission before retention. Every memory claim starts with an admission deck. Cue-time readout must verify that the cue is visible when it appears. Shortcut checks must show that endpoint images, action-only traces, and state-only traces do not solve the label. Counterfactual masks must verify that post-cue frames do not leak the answer. Only after these checks pass do we run retention, exposure, or executed-use tests. This staging is conservative, but it is what makes a negative or mixed result interpretable rather than merely disappointing.

3 Environments and claim boundaries

Figure 2 is the main evidence ledger. The rows are not pooled into a single score because each environment is serving a different purpose. PushT tests whether a visual-binding cue can influence an executed decision on two frozen host families. PointMaze tests whether retained goal evidence can drive a navigation consumer. Wall is included specifically to expose a bottleneck where carrier storage survives but host exposure fails. OGBench supplies breadth for the cue-admission machinery across navigation, locomotion, manipulation, scene, and puzzle families. Its repaired native-use stage is reported as fixed-controller success for audited PointMaze/Cube/Puzzle/Scene controllers, not as native world-model planning or autonomous memory discovery.

This ledger is the central presentation choice. A conventional results table would invite a misleading reading in which environments are interchangeable and carrier names can be ranked globally. The audit instead asks where the old evidence enters, where it disappears, and whether disappearance occurs before or after the frozen host.

4 Positive use cases

The strongest evidence in this study is not a probe, but executed use. Figure 3 compares host-readability and executed success under fixed consumers. The main observation is qualitative: when the host-exposed state is reliable, a simple consumer can turn old evidence into control; when the controls remove pre-context evidence, execution drops to the expected uninformed regime.

PushT: use is not host-family-specific. The PushT binding deck asks whether a transient visual token can be retained until a later action choice. The result passes on both DINO-WM and LeWM frozen hosts under the same style of full/reset/no-state comparison. This matters because a single-host success could be dismissed as a quirk of one feature space. The evidence here is narrower and cleaner: two host families can expose the same kind of old visual binding strongly enough for a fixed consumer to act on it. The claim remains bounded because the banks are host-native rather than a fully row-matched simulator comparison.

PointMaze: retained state can become navigation. PointMaze is the clearest use example because the old cue selects among waypoint goals and the downstream controller is fixed.

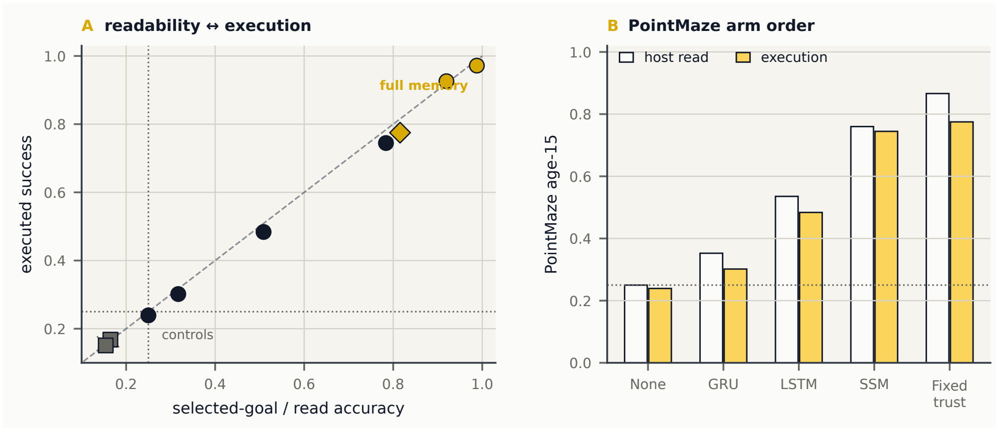

Figure 3: Readability transfers to execution when the consumer is fixed. PushT shows a checkpointed visual-binding use gate on DINO-WM and LeWM. PointMaze shows the clearest large execution separation: memory-conditioned goal selection succeeds where the no-state consumer behaves like an uninformed policy.

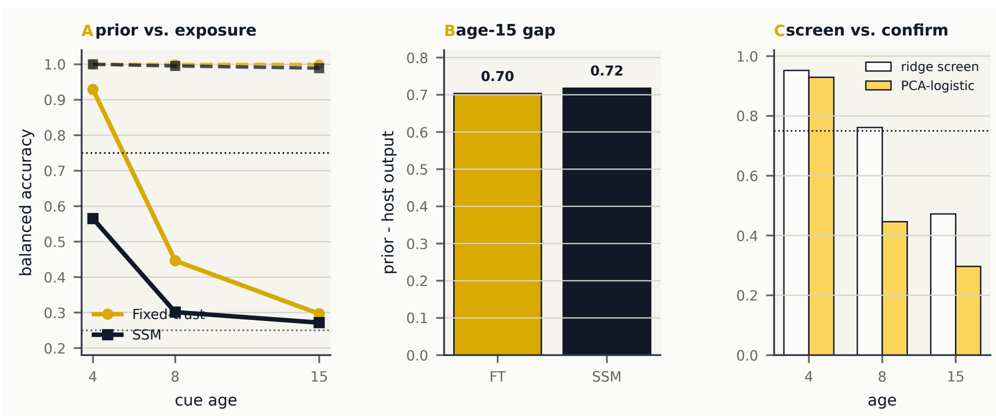

Figure 4: Wall localizes the bottleneck. Dashed curves in panel A are carrier-prior readouts; solid curves are host-output readouts. The widening gap shows that retaining a cue internally is not the same as exposing it through the frozen predictor. Panel C shows why the paper uses the stricter PCA-logistic confirmation instead of the earlier ridge screen.

The memory sidecar does not receive a privileged goal label at decision time; it must carry the transient mini-map evidence until the consumer reads the host output. The large separation between full memory and no-state execution shows that the retained evidence is not merely decodable; it changes the chosen path under a pinned controller. At the same time, this is not native DINO-WM planning. It is evidence that a fixed external consumer can use exposed memory.

5 Failure localization: storage is not exposure

Wall is the most important negative result. It was added to answer a question that positive execution decks cannot answer: if a model fails at long evidence age, did the side state forget, or did the frozen host stop exposing what the side state still contains? Figure 4 shows that these are separable. The carrier-prior readout remains high across ages, while the host-output readout decays toward the shortcut regime. The stricter PCA-logistic confirmation also reverses an overly optimistic screening interpretation: Wall should be written as a bottleneck diagnosis, not a success case.

This result is useful precisely because it is not flattering. It rules out a simple story in which the proposed sidecar is a universally better memory module. Instead, the evidence points to a host-interface failure: old information can survive in the carrier while the frozen predictor

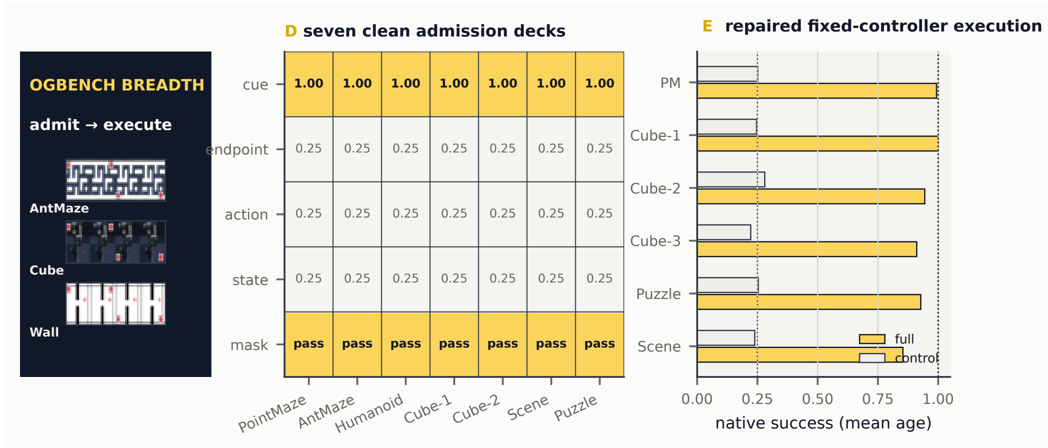

Figure 5: OGBench breadth. Cue labels are readable at cue time, while endpoint, action, and state shortcuts remain at the uninformed baseline; counterfactual masks pass in all seven decks. The execution panel summarizes the repaired fixed-controller native-use sweep: full memory converts the selected target into native success on the audited PointMaze/Cube/Puzzle/Scene rows, while recent-only and random target selection remain near chance. This is fixed-controller use evidence, not native planner evidence.

maps the legal context to an output that no longer carries the needed variable. A paper that reports only carrier-prior accuracy would miss this failure.

6 Protocol breadth: OGBench admission and native-use checks

OGBench (Park et al., 2025) is used as an admission bridge. We construct cue-only counterfactual decks for seven render-ready environments: PointMaze, AntMaze, HumanoidMaze, Cube-single, Cube-double, Scene, and Puzzle. The purpose is not to claim that an OGBench world model remembers. The purpose is to test whether the same cue protocol can create valid old-evidence targets without shortcut leakage across qualitatively different visual domains. We then run a deliberately bounded feature-host follow-up, then convert the selected memory target into native environment success wherever a local audited controller exists. The repaired all-env native-use sweep scores 24/36 env-age rows across non-teleport PointMaze, Cube-single/double/triple, Puzzle-3x3, and Scene. PointMaze uses a BFS waypoint controller in the native point-mass action space. Cube variants use the OGBench CubeMarkovOracle with live target metadata and multi-block ordering. Puzzle uses a lights-out solve with the button oracle, and Scene composes the button, drawer, window, and cube Markov oracles with dependency-aware sequencing. PointMaze-teleport is left unscored because its oracle-label controller gate reaches only 0.608–0.658 below the 0.850 admission threshold; AntMaze and HumanoidMaze are left unscored because this repository has no audited low-level controller for them. This tests whether recovered memory changes final task success; it does not test autonomous cue discovery or native OGBench planning.

The OGBench native-use result supplies the straightforward performance number missing from a pure readout study. Across the 24 scored env-age rows, full memory reaches mean native success 0.952, with worst scored success 0.828 and mean full-minus-control lifts 0.714 over recent-only and 0.708 over random selection. The scientific value is not that the fixed controllers are new; they are deliberately simple. The value is that the same memory-selected label that is readable at the host interface also determines whether the environment reports success. This fixed-controller stage still supplies the old cue feature to the sidecar; it is therefore evidence of use, not evidence of autonomous evidence discovery.

Removing the manual cue-feature handoff. We therefore run a final non-manual OGBench stage. The learning rule sees only the rendered causal stream, action tokens, and time tokens. It mines salient target frames and patch sets automatically, masks the selected target view, and predicts fixed generic color/texture patch latents from a compact slot bottleneck. The target is not a cue crop, the model is not told which moment is key, and the training loss

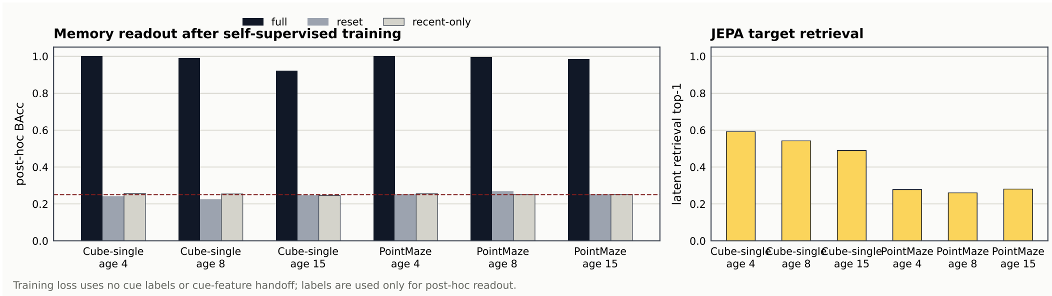

Figure 6: Final non-manual OGBench result. The salient patch-set JEPA objective passes all 30 registered cells over two environments, three cue ages, and five seeds. Full-memory readout remains high at age 15, while reset and recent-only controls stay near the four-way uninformed baseline.

Table 1: Final non-manual OGBench summary. Gate: full balanced accuracy at least 0.750 and reset/recent-only controls at most 0.350 for every seed.

Deck	Age	Seeds	Full	Reset	Recent-only	Status
Cube-single	4	5/5	1.000	0.240	0.257	✓
Cube-single	8	5/5	0.990	0.223	0.254	✓
Cube-single	15	5/5	0.921	0.247	0.246	✓
PointMaze-large	4	5/5	1.000	0.251	0.255	✓
PointMaze-large	8	5/5	0.994	0.266	0.250	✓
PointMaze-large	15	5/5	0.983	0.251	0.252	✓

never uses cue labels. The labels are used only after training to audit whether the learned slot state retains the old cue under full/reset/recent-only contrasts.

Breadth stress tests. The two-deck result above is the current clean paper-level non-manual claim. The broader PointMaze/Cube/Scene/Puzzle/locomotion sweep is treated as a stress test rather than an automatic headline. Its generated status is recorded in Appendix C; rows that require large-bank confirmation are reported as such instead of being folded into a single pooled score.

7 Cross-study interpretation

The interface chain, not the carrier name, is the unit of evidence. The same sidecar can look strong under a carrier-prior readout and weak under host-output exposure. Conversely, a modest readout can still matter if a consumer is sensitive to the recovered variable. This is why the paper avoids phrases such as "fixed-trust solves memory" or "state-space is the memory winner." The defensible statement is conditional: for a given target, host, sidecar, and consumer, the old evidence either survives the chain or it does not.

Negative results sharpen the method. The Wall result is not dead weight; it is the diagnostic that makes the positive PushT and PointMaze results credible. Without Wall, the paper could be read as another architecture-comparison story. With Wall, the paper shows why architecture comparisons are brittle: a carrier can preserve the symbol while the host interface erases it. That is the phenomenon reviewers should remember.

Execution should be interpreted as consumer-specific use. The fixed consumers are deliberately simple and shared within each deck. They prove that exposed memory can change control, not that the native planner has learned to query memory. This distinction is not a weakness to hide. It is the correct boundary between an interface audit and an end-to-end planning paper.

Breadth should not outrun evidence. The OGBench decks show that the admission stage generalizes beyond the original hosts and tasks, and the expanded fixed-controller use stage shows that recovered memory can change native environment success across audited Point-

Table 2: How to read audit signatures. The entries are written as diagnoses, not as leaderboard ranks.

Observed signature	Interpretation	Correct intervention target
Prior weak, host output weak, controls at chance	The side state did not preserve the old evidence in a useful form.	Memory objective, write dynamics, capacity, or training horizon.
Prior strong, host output weak	The carrier stored evidence, but the frozen fusion/predictor path did not expose it.	Exposure adapter, fusion point, residual coordinates, or host-aware auxiliary objective.
Host output readable, execution unchanged	The representation is accessible to a probe but not used by the declared consumer.	Consumer interface, action grounding, or planner integration.
Full beats no state but not reset	The sidecar helps the endpoint, but dependence on pre-context state is not isolated under the reset intervention.	Report as sensitivity, not persistent retention; redesign reset or task.
Admission passes but no host stage exists	The task is clean, not yet a memory experiment.	Add a frozen host, carrier grid, reset/no-state controls, and use deck.

Maze/Cube/Puzzle/Scene rows. Treating this as native planner memory would inflate the paper but weaken its rigor. The non-manual patch-set JEPA stage takes the next step by removing the explicit cue-feature handoff on two core OGBench environments. The right boundary is therefore: autonomous old-evidence retention is supported for PointMaze-large/Cube-single render-stage memory, while native OGBench world-model planning and broad OGBench-wide autonomous generalization remain unclaimed.

8 Diagnostic consequences

The audit is useful only if its outcomes imply different next actions. Table 2 summarizes the signatures exposed by the current suite. The same endpoint score can have different causes depending on the carrier prior, reset, no-state, host output, and execution views.

This decision table clarifies why Wall is valuable. A positive-only paper would treat Wall as a failed environment and move on. A memory audit treats it as evidence about the interface. The prior is strong enough to rule out simple forgetting inside the fixed-trust carrier; the host output is weak enough to rule out successful exposure. That points to a concrete repair hypothesis: a self-supervised memory method should not only store masked evidence, but also train the host-facing projection to keep that evidence in coordinates that survive frozen predictor mixing.

9 Environment roles and why more rows are not automatically better

The current environments are not interchangeable. PushT, PointMaze, Wall, and OGBench each stress a different part of the ladder. Table 3 states the role of each environment in the argument and keeps the paper from reading like an experiment log.

This structure is stronger than adding many partially completed environments. The paper now spans positive use, exposure failure, protocol breadth, and fixed-controller native

Table 3: Environment roles in the paper narrative.

Environment	What it contributes	What it does not prove	Why it belongs in main text
PushT	Visual binding under two frozen host families with an executed-use gate.	A row-matched causal comparison between DINO-WM and LeWM.	Shows the use gate is not host-family specific.
PointMaze	Goal memory that survives host readout and changes navigation under a fixed controller.	Native model planning with memory.	Provides the strongest positive behavioral link.
Wall	A controlled negative case where storage and exposure separate.	Broad success of the proposed carrier.	Makes the diagnostic thesis falsifiable and prevents overclaiming.
OGBench	Seven clean cue decks, plus repaired PointMaze/Cube/Puzzle/Scene native success under fixed controllers after memory-selected target recovery, plus a non-manual salient patch-set JEPA stage on PointMaze-large/Cube-single.	Native OGBench world-model planning or broad OGBench-wide autonomous generalization.	Gives a reviewer-facing final-performance check and a clean autonomous memory-learning result beyond linear readout.

success. The missing piece is not another admission deck; it is an autonomous evidence-discovery stage and audited locomotion controllers. AntMaze remains the most useful next execution target for locomotion generalization, but it requires a real ant controller or policy before it can be promoted beyond feature-host capacity.

10 Implications for memory-module design

The results suggest a design principle for future modules: self-supervised memory should optimize an evidence interface, not merely a hidden state. A module that reconstructs or predicts masked old evidence internally can still fail if its host-facing projection is not trained to expose that evidence. A module that exposes evidence can still fail if the consumer reads the wrong coordinates. Therefore the objective should couple three pressures: retain masked evidence without an explicit key-frame label, expose it through the frozen host interface, and preserve downstream separability under reset and no-state controls.

This principle is compatible with JEPA-style training. The sidecar need not decode pixels or be told which observation is important for the final decision. Instead, the final OGBench stage mines salient target frames and patch sets without labels, predicts fixed generic color/texture latents, and audits the result only after training. The important lesson from Wall is that this target should remain host-facing: optimizing only the carrier prior risks building a memory that exists in the side state but is invisible after frozen predictor mixing. The important lesson from PointMaze and Cube is that a simple self-supervised patch-set target can be sufficient when the retained coordinates preserve the old evidence under reset and recent-only controls.

For paper framing, this means the study should not sell fixed-trust as the new winning architecture. The stronger claim is that the audit identifies the right unit of design—the evidence interface—and that salient patch-set JEPA is one concrete non-manual instantiation that passes a registered PointMaze/Cube grid. The remaining claim boundary is breadth and native planning: the method has not yet been certified across every admitted OGBench deck or inside an end-to-end world-model planner.

11 Related work and positioning

Recent work adds memory to world models through recurrent or state-space agents, transformer memory, explicit retrieval, spatial/geometric memory, and world-action memory systems (Samsami et al., 2024; Laird & Clark, 2025; Deng et al., 2023; Zheng et al., 2026; Yang et al., 2026; Wang et al., 2026; Garcin et al., 2026; Lu et al., 2026). These papers primarily ask how to build stronger memory-enabled agents or simulators, often evaluating through return, rollout consistency, video reconstruction, or task success. This paper asks a different question: under a frozen finite-context latent host, which interface rung actually carries the old evidence?

The closest host background is DINO-WM (Zhou et al., 2025) and LeWorldModel (Maes et al., 2026). We freeze these hosts and treat memory as a side-interface audit rather than a new host training recipe. This positioning is also consistent with broader cautions about probing and planning: decodability does not by itself establish task use (Belinkov, 2022; Ravichander et al., 2021), and predictive feature quality does not guarantee downstream planning utility (Li et al., 2026).

12 Limitations

The main limitations are claim-boundary limitations. First, the paper is not a universal architecture win. Wall shows that an internally strong carrier can fail at host exposure, and the claim ledger should keep this visible. Second, the strongest execution results use fixed external consumers and pinned controllers, not native planners; PointMaze-teleport fails the controller gate and Ant/Humanoid locomotion remains unscored without a local low-level controller. Third, the non-manual patch-set JEPA result is currently certified on PointMaze-large and Cube-single, not across every admitted OGBench deck. Fourth, OGBench still does not include a native world-model planner that autonomously queries the learned memory. Fifth, the PushT cross-host result establishes that the use gate can pass on two host families, but it is not a row-matched causal comparison between those families. These limitations are not cosmetic; they define what the audit can and cannot certify.

13 Conclusion

The evidence supports a concise ICLR story: persistent memory in finite-context latent world models is an interface property. The side state may retain old evidence, the frozen host may expose or suppress it, and a consumer can use it only when the exposed representation remains reliable. PushT and PointMaze provide positive executed-use cases; Wall identifies the storage-versus-exposure bottleneck; OGBench shows that valid old-evidence admission decks can be carried through to repaired 24/36 env-age fixed-controller native-use success and that a non-manual salient patch-set JEPA objective can retain old evidence on PointMaze-large and Cube-single without cue labels, cue crops, or key-frame annotations. The resulting claim is deliberately scoped: the audit can certify where evidence survives the interface chain, but it does not turn any single carrier name into a universal memory solution.

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A Artifact map

The paper is generated from current local artifacts rather than hand-entered plots. The figure-generation script is scripts/plot_paper_b.py. It writes figures to paper_b/figures/ and a JSON snapshot to paper_b/generated_results/result_snapshot.json.

Table 4: Primary local result sources used by this paper.

Claim	Local source
PushT checkpointed downstream use	outputs/pusht_checkpointed_downstream_use_v1/summary.json
PointMaze carrier grid	outputs/dinowm_pointmaze_wave3/formal/carrier_summary.json
PointMaze external use	outputs/dinowm_pointmaze_wave3/formal/external_use_summary.json
Wall PCA-logistic confirmation	outputs/dinowm_wall_audit_v1/stage_h_logistic_readout/summary.json
Wall ridge screen	outputs/dinowm_wall_audit_v1/stage_h_carriers/summary.json
OGBench admission bridge	outputs/ogbench_mesmem_admission_v1/summary_all.json
OGBench feature-host capacity	outputs/ogbench_feature_host_stage_v1/summary.json
OGBench native-use success	outputs/ogbench_native_use_stage_v1/summary.json
OGBench repaired all-env native-use success	outputs/native_use_repaired_combined_v1/summary.json
Final non-manual OGBench patch-set JEPA	outputs/random_patchset_view_color_jepa_final_v1/summary.json

B Generated all-env native-use repair monitor

This section is generated from outputs/native_use_repaired_combined_v1/summary.json. It records the fixed-controller native-use sweep after the multi-object controller repair. Rows without an audited local controller, or rows whose oracle-label controller gate fails, remain unscored scope limits.

Table 5: Repaired all-env fixed-controller native-use summary. Full memory is the ME-JEPA memory-selected target; control is the stronger of recent-only and random selection, averaged over the listed rows. The sweep scores 24/36 env-age rows, with mean full success 0.952, worst scored full success 0.828, and mean full-minus-control lifts 0.714 over recent-only and 0.708 over random.

Group	Rows	Full	Control	Oracle	Status
PointMaze non-teleport	9	0.993	0.252	1.000	scored
Cube-single	3	1.000	0.246	1.000	scored
Cube-double	3	0.944	0.281	0.944	repaired
Cube-triple	3	0.911	0.222	0.911	repaired
Puzzle-3x3	3	0.928	0.254	1.000	scored
Scene	3	0.854	0.239	0.903	repaired
PointMaze-teleport	0	—	—	0.608–0.658	gate failed
Ant/Humanoid locomotion	0	—	—	—	no local controller

C Generated non-manual breadth monitor

This section is generated automatically by the overnight controller. It is included to remove ambiguity between clean paper-level claims and broader stress tests. A row marked mixed is not promoted to a main-text claim; it identifies an environment-age setting that requires confirmation or a design change.

Generation status. Overnight sequence completed. Last update: 2026-07-17T02:57:48.

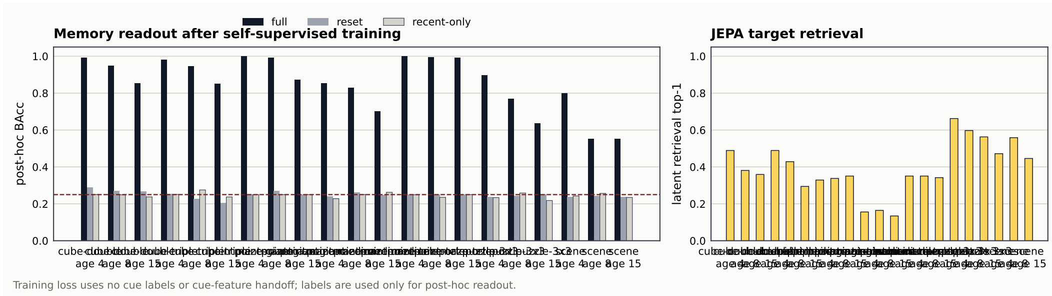

Figure 7: Generated non-manual breadth plot for Random patch-set breadth wave.

Table 6: Generated non-manual breadth table for Random patch-set breadth wave.

Deck	Age	Seeds	Pass	Full	Reset	Recent	Status
cube-double-play-v0	4	3	3/3	0.991	0.289	0.250	pass
cube-double-play-v0	8	3	2/3	0.949	0.270	0.249	mixed
cube-double-play-v0	15	3	3/3	0.854	0.266	0.238	pass
cube-triple-play-v0	4	3	3/3	0.981	0.255	0.251	pass
cube-triple-play-v0	8	3	3/3	0.947	0.228	0.275	pass
cube-triple-play-v0	15	3	3/3	0.852	0.206	0.237	pass
pointmaze-giant-navigate-v0	4	3	3/3	1.000	0.250	0.250	pass
pointmaze-giant-navigate-v0	8	3	3/3	0.993	0.271	0.250	pass
pointmaze-giant-navigate-v0	15	3	3/3	0.874	0.250	0.250	pass
pointmaze-medium-navigate-v0	4	3	3/3	0.853	0.240	0.228	pass
pointmaze-medium-navigate-v0	8	3	2/3	0.830	0.262	0.251	mixed
pointmaze-medium-navigate-v0	15	3	1/3	0.702	0.249	0.263	mixed
pointmaze-teleport-navigate-v0	4	3	3/3	1.000	0.250	0.249	pass
pointmaze-teleport-navigate-v0	8	3	3/3	0.996	0.250	0.235	pass
pointmaze-teleport-navigate-v0	15	3	3/3	0.993	0.250	0.250	pass
puzzle-3x3-play-v0	4	3	3/3	0.896	0.238	0.234	pass
puzzle-3x3-play-v0	8	3	1/3	0.770	0.242	0.259	mixed
puzzle-3x3-play-v0	15	3	1/3	0.636	0.250	0.218	mixed
scene-play-v0	4	3	2/3	0.799	0.238	0.243	mixed
scene-play-v0	8	3	0/3	0.553	0.243	0.257	mixed
scene-play-v0	15	3	0/3	0.552	0.239	0.235	mixed

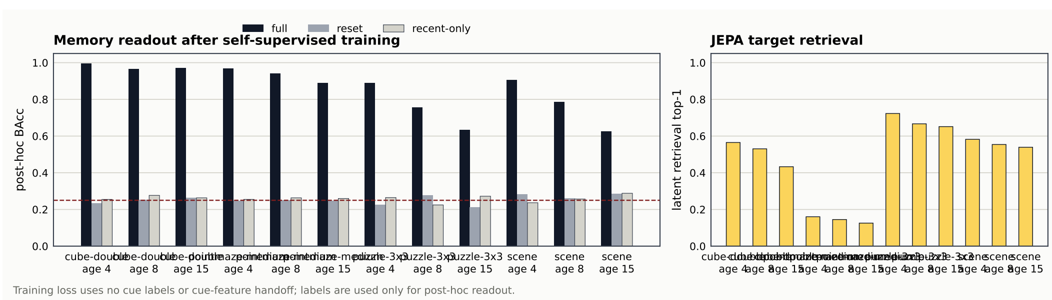

Figure 8: Generated non-manual breadth plot for Large-bank confirmation.

Random patch-set breadth wave. This generated block reports 13/21 environment-age rows passing the registered non-manual gate. Mixed rows are treated as stress-test limits or confirmation targets, not as headline wins.

Large-bank confirmation. This generated block reports 8/12 environment-age rows passing the registered non-manual gate. Mixed rows are treated as stress-test limits or confirmation targets, not as headline wins.

antmaze-large-navigate-v0. This generated block reports 3/3 environment-age rows passing the registered non-manual gate. Mixed rows are treated as stress-test limits or confirmation targets, not as headline wins.

antmaze-giant-navigate-v0. This generated block reports 3/3 environment-age rows passing the registered non-manual gate. Mixed rows are treated as stress-test limits or confirmation targets, not as headline wins.

Table 7: Generated non-manual breadth table for Large-bank confirmation.

Deck	Age	Seeds	Pass	Full	Reset	Recent	Status
cube-double-play-v0	4	3	3/3	0.995	0.233	0.255	pass
cube-double-play-v0	8	3	3/3	0.964	0.253	0.277	pass
cube-double-play-v0	15	3	3/3	0.969	0.264	0.263	pass
pointmaze-medium-navigate-v0	4	3	3/3	0.968	0.248	0.255	pass
pointmaze-medium-navigate-v0	8	3	3/3	0.941	0.249	0.263	pass
pointmaze-medium-navigate-v0	15	3	3/3	0.888	0.246	0.260	pass
puzzle-3x3-play-v0	4	3	3/3	0.888	0.225	0.265	pass
puzzle-3x3-play-v0	8	3	1/3	0.757	0.277	0.225	mixed
puzzle-3x3-play-v0	15	3	0/3	0.632	0.212	0.272	mixed
scene-play-v0	4	3	3/3	0.906	0.282	0.237	pass
scene-play-v0	8	3	1/3	0.787	0.260	0.257	mixed
scene-play-v0	15	3	0/3	0.626	0.286	0.288	mixed

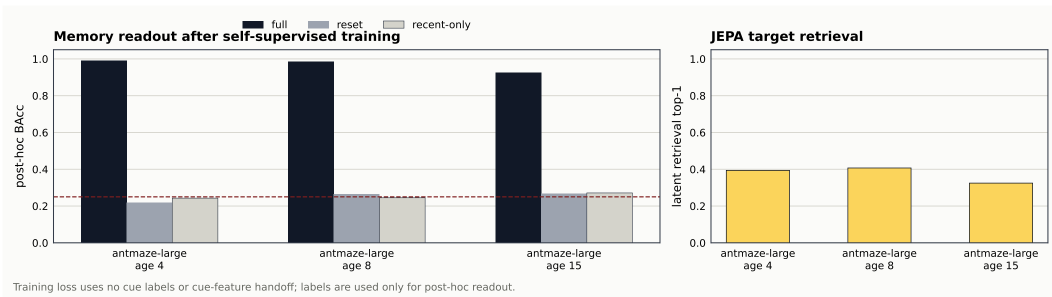

Figure 9: Generated non-manual breadth plot for antmaze-large-navigate-v0.

Table 8: Generated non-manual breadth table for antmaze-large-navigate-v0.

Deck	Age	Seeds	Pass	Full	Reset	Recent	Status
antmaze-large-navigate-v0	4	3	3/3	0.992	0.220	0.243	pass
antmaze-large-navigate-v0	8	3	3/3	0.985	0.264	0.245	pass
antmaze-large-navigate-v0	15	3	3/3	0.925	0.268	0.271	pass

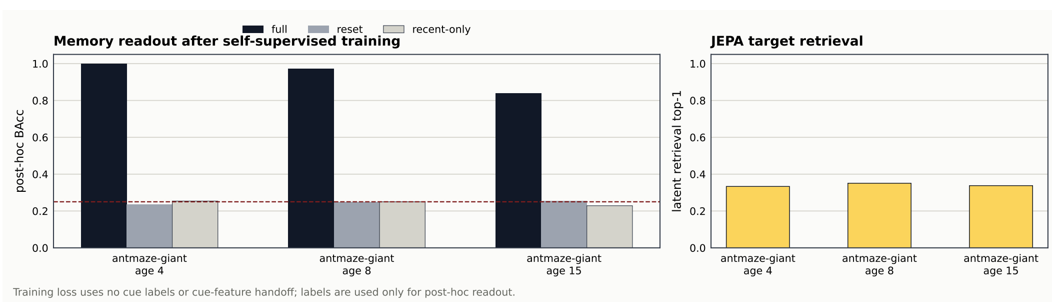

Figure 10: Generated non-manual breadth plot for antmaze-giant-navigate-v0.

Table 9: Generated non-manual breadth table for antmaze-giant-navigate-v0.

Deck	Age	Seeds	Pass	Full	Reset	Recent	Status
antmaze-giant-navigate-v0	4	3	3/3	1.000	0.235	0.254	pass
antmaze-giant-navigate-v0	8	3	3/3	0.972	0.250	0.250	pass
antmaze-giant-navigate-v0	15	3	3/3	0.839	0.255	0.229	pass

Table 10: Generated non-manual breadth table for humanoidmaze-large-navigate-v0.

Deck	Age	Seeds	Pass	Full	Reset	Recent	Status
humanoidmaze-large-navigate-v0	4	3	1/3	0.626	0.234	0.247	mixed
humanoidmaze-large-navigate-v0	8	3	0/3	0.585	0.236	0.250	mixed
humanoidmaze-large-navigate-v0	15	3	0/3	0.476	0.285	0.236	mixed

humanoidmaze-large-navigate-v0. This generated block reports 0/3 environment-age rows passing the registered non-manual gate. Mixed rows are treated as stress-test limits or confirmation targets, not as headline wins.

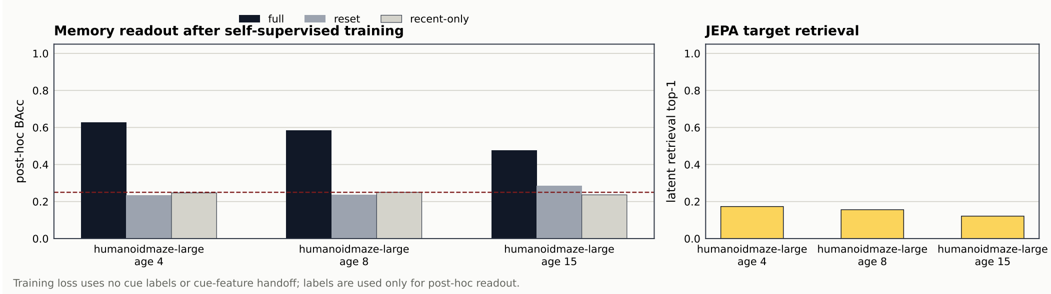

Figure 11: Generated non-manual breadth plot for humanoidmaze-large-navigate-v0.

D Detailed claim ledger

Table 11: Expanded ledger corresponding to Figure 2. The main paper uses the visual ledger; this table records the detailed boundary for each row.

Environment	Host / task	Retention	Exposure	Use	Boundary
PushT	DINO-WM and LeWM, six-way visual binding, age 15	✓	✓	✓	Cross-host use gate passes; host-native banks prevent a row-matched causal comparison between host families.
PointMaze	DINO-WM, four-goal transient cue	✓	✓	✓	Strong external consumer result; not native planner memory.
Wall	DINO-WM, four-position transient wall cue	✓	mixed	—	Carrier-prior memory remains high; host-output exposure confirms only at short cue age.
OGBench	Seven admitted decks; repaired fixed-controller native use plus two-deck non-manual patch-set JEPA	✓	✓	✓	Admission passes across seven decks. The repaired native-use sweep scores 24/36 env-age rows under fixed controllers, and the non-manual patch-set JEPA stage passes all 30 registered PointMaze-large/Cube-single retention cells. No native OGBench planner is claimed.

E Quantitative result summary

PushT checkpointed use. At age 15, both frozen host families support the same qualitative conclusion: full memory separates from reset and no-state controls in host readout and executed success. DINO-WM full memory reaches balanced accuracy 0.988 and executed success 0.972, while reset/no-state execution remains near the six-way uninformed regime. LeWM full memory reaches balanced accuracy 0.920 and executed success 0.926 with the same control pattern.

Table 12: PushT binding use summary.

Host	Full execution	Reset execution	No-state execution
DINO-WM	0.972	0.167	0.168
LeWM	0.926	0.156	0.151

PointMaze external use. The fixed consumer preserves the carrier ordering in navigation execution. No state remains close to random-goal behavior; state-space and fixed-trust are the strongest arms.

Table 13: PointMaze external-use summary.

Arm	Goal/read accuracy	Executed success
No state	0.250	0.239
GRU	—	0.302
LSTM	—	0.484
State-space	0.760	0.745
Fixed-trust	0.866	0.775

Wall readout note. The Wall carrier grid was first screened with a deterministic ridge readout. That screen suggested fixed-trust might clear host-output exposure at ages 4 and 8. The paper uses the stricter confirmation run instead: StandardScaler, PCA(256), and logistic regression fit only on the training split. Under this confirmation fixed-trust clears only age 4, while state-space does not clear any age. This is why Wall is described as exposure failure localization rather than a broad success result.

Table 14: Wall confirmation: carrier prior remains high while host-output exposure decays.

Arm	Readout	Age 4	Age 8	Age 15
Fixed-trust	carrier prior	1.000	1.000	1.000
Fixed-trust	host output	0.929	0.446	0.297
State-space	carrier prior	1.000	0.995	0.989
State-space	host output	0.565	0.301	0.272

F OGBench claim boundary

The OGBench admission bridge checks only whether a controlled cue deck is valid: cue-time encoding must be high, endpoint/action/state shortcuts must be near chance, and counterfactual masks must not leak post-cue evidence. It does not train a frozen OGBench world model and does not run downstream execution.

The follow-up feature-host stage is one rung stronger but still bounded. It uses admitted decks, DINOv2 pooled patch features as a frozen feature interface, and a sidecar that carries the old cue feature before writing a residual into the endpoint feature. The writer loss is self-supervised against the cue feature, while labels are used only for the post-hoc readout.

The native-use stage then asks whether the selected label changes final environment success. Non-teleport PointMaze uses a BFS waypoint controller over native point-mass actions. Cube variants use the OGBench CubeMarkovOracle with selected-label target metadata, Puzzle uses a lights-out solve with the button oracle, and Scene composes button, drawer, window, and cube Markov oracles. The repaired sweep scores 24/36 env-age rows; the PointMaze-teleport controller gate remains below threshold, and AntMaze/HumanoidMaze remain unscored without local low-level controllers. These are fixed-controller use checks; they are not native OGBench world-model planning.

The final non-manual stage then removes the explicit cue-feature handoff on PointMaze-large and Cube-single. It mines salient target frames and patch sets without cue labels or cue-layout knowledge, predicts fixed generic color/texture patch latents through compact slots, and evaluates the learned state only after training. This certifies autonomous old-evidence retention for the two core render-stage decks, but not yet broad OGBench-wide generalization or native planner memory.

Table 15: OGBench admission summary. All rows pass cue-time readability, shortcut, and mask checks; all memory/use columns are intentionally unclaimed.

Deck	Family	Cue read	Max shortcut	Mask audit
PointMaze-large-navigate	navigation	1.000	0.250	pass
AntMaze-large-navigate	locomotion	1.000	0.250	pass
HumanoidMaze-large-navigate	high-DoF locomotion	1.000	0.250	pass
Cube-single-play	object manipulation	1.000	0.250	pass
Cube-double-play	object manipulation	1.000	0.250	pass
Scene-play	multi-object scene	1.000	0.250	pass
Puzzle-3x3-play	combinatorial puzzle	1.000	0.250	pass

Table 16: OGBench feature-host capacity summary. Gate: full host-output balanced accuracy at least 0.750, reset and no-state controls at most 0.300. All rows use 200 train bases, 80 validation bases, and five seeds.

Deck	Age	Seeds passing	Full	Reset/no-state
AntMaze-large-navigate	4	5/5	1.000	0.250 / 0.250
AntMaze-large-navigate	8	5/5	1.000	0.250 / 0.250
AntMaze-large-navigate	15	5/5	1.000	0.250 / 0.250
Cube-single-play	4	5/5	1.000	0.250 / 0.250
Cube-single-play	8	5/5	1.000	0.250 / 0.250
Cube-single-play	15	5/5	1.000	0.250 / 0.250

Table 5 reports the repaired all-env native-use sweep. It subsumes the earlier PointMaze-large/Cube-single slice and keeps controller-gate failures and unavailable locomotion controllers as explicit scope limits.

Table 17: Final non-manual OGBench patch-set JEPA summary. Labels are used only for post-hoc audit; training uses no cue labels, cue crops, or key-frame annotations.

Deck	Age	Seeds passing	Full	Reset	Recent-only
PointMaze-large-navigate	4	5/5	1.000	0.251	0.255
PointMaze-large-navigate	8	5/5	0.994	0.266	0.250
PointMaze-large-navigate	15	5/5	0.983	0.251	0.252
Cube-single-play	4	5/5	1.000	0.240	0.257
Cube-single-play	8	5/5	0.990	0.223	0.254
Cube-single-play	15	5/5	0.921	0.247	0.246